BETOBULLET

10 HP · EURORACK FORMAT
FOR VCV RACK 2



BURN

DUAL FUNCTION GENERATOR WITH VCAS

USER MANUAL · v2.0.0 · October 2026

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Two envelope/LFO channels with a VCA each: punchy percussive decays, gate-held swells, slew for any CV, loops from minutes long to audio rate, end-of-cycle pulses and a sum output.

1. INTRODUCTION

Burn packs two complete function generators into 10 HP. Each channel is a slew limiter driven by a pulse generator: a trigger produces an attack–decay envelope, a gate at the slew input produces an attack–sustain–release envelope, and any CV at the slew input comes out slewed. The LFO switch loops the envelope, from cycles of more than an hour to audio rate. A VCA on every channel shapes either the envelope itself or any signal you patch into it.

Burn recreates the published behaviour of the Cosmotronic Delta-V, an all-analogue module. The panel follows the lettering of the original manual (A–P).

INSTALLATION

Copy `Burn-2.0.0-win-x64.vcvplugin` to `Documents/Rack2/plugins` and restart VCV Rack. The module is listed under the brand **betobullet**.

2. PANEL OVERVIEW



A **Attack time**
Rise time, 0.3 ms to about 40 s. The tooltip shows the real time.

B **Decay time**
Fall time, 0.3 ms to about 40 s.

C **Slope shape**
Curve of both segments: exponential, linear or logarithmic.

D **One-shot / LFO switch**
In LFO mode the envelope retriggers itself at the end of every cycle.

E **Output attenuverter**
Level of the channel output, from −100 % to +100 % (±10 V).

F **Trigger link**
Connects trigger input G to channel 2 (G > J).

G **Trigger input**
Any rising signal fires an attack–decay envelope on channel 1.

H **Slew input**
A gate makes an ASR envelope; any CV comes out slewed.

I **VCA in, channel 1**
Normalled to 5 V: with nothing patched the output is the envelope.

J **Trigger input, channel 2**
Also fired by G when the link switch is on.

K **CV in for attack**
+1 V halves the attack time.

L **CV in for decay**
+1 V halves the decay time.

M **VCA in, channel 2**
Normalled to 5 V.

N **Channel 1 output**
Envelope × VCA × attenuverter.

O **End of cycle out**
A 0.5 ms trigger when the cycle has finished.

P **Sum out of 1 + 2**
Both channel outputs added.

3. OPERATION

The controls of channel 2 mirror those of channel 1 on the right half of the panel.

TRIGGER INPUT

Whatever you patch into the trigger input **(G)**, its rising edge launches one attack–decay cycle. While the envelope is still **rising**, further triggers are simply ignored; once it is **falling**, a trigger sends it back up from wherever it is. Long attacks therefore swallow some of the incoming pulses, which turns a clock into divisions and irregular rhythms.

SLEW INPUT

Feed a gate into the slew input **(H)** and the output climbs at the attack rate, stays up for as long as the gate is held and then falls at the decay rate: an attack–sustain–release shape. Any other CV patched here comes out slew-limited, rising at the attack rate and falling at the decay rate.

VCA

Each VCA input **(I, M)** sees a fixed 5 V when nothing is plugged in, so the output is simply the envelope at the level of the attenuverter **(E)**. Plug in audio and the envelope now opens and closes that sound. Plug in a stepped CV and every envelope comes out at a different height: an easy way to add accents.

LFO SWITCH

Flip the LFO switch **(D)** and the channel retriggers itself every time it reaches the bottom. The loop runs from audio rate (about 1.6 kHz) down to glacial: negative CV on the time inputs stretches a single cycle past twenty minutes.

END OF CYCLE

The end-of-cycle output **(O)** fires a short pulse the moment a channel lands back at zero, once per loop in LFO mode.

TRIGGER LINK

With the link switch **(F)** up, trigger 1 drives channel 2 too, so one cable fires both envelopes.

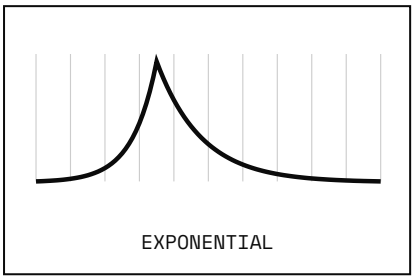
SUM OUTPUT

The sum output **(P)** adds both channel outputs. With opposite attenuverter settings it makes bipolar shapes.

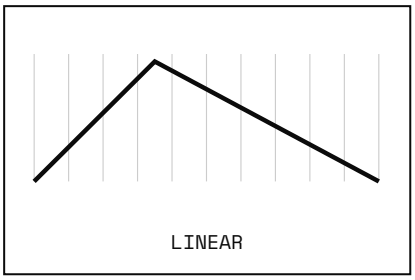
4. RESEARCH DATA

Responses measured on Burn (48 kHz, linear shape unless noted).

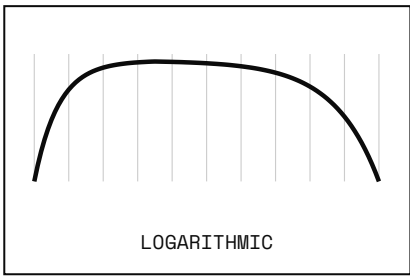
ENVELOPE SHAPES



EXPONENTIAL

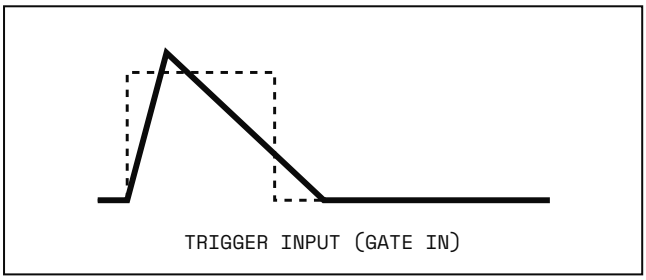


LINEAR

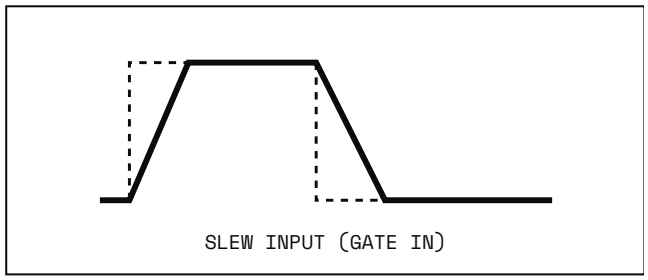


LOGARITHMIC

TRIGGER AND SLEW INPUTS



TRIGGER INPUT (GATE IN)



SLEW INPUT (GATE IN)

QUANTITY	VALUE
Max output	+10 V to −10 V, soft limited
Fastest rise / fall	0.3 ms each
Max LFO frequency	about 1.6 kHz
Slowest stage	about 40 s at the knob, over 20 min with −5 V of CV
EOC trigger width	0.5 ms
Time CV	+1 V halves the time (exponential)

5. PATCH IDEAS

- **Kick drum**: a clock into G, attack minimum, decay around 150 ms, shape exponential, a sine VCO into I.
- **Clock divider**: a long attack ignores the triggers that fall inside it. Take EOC as the divided clock.
- **Accented envelopes**: a stepped sequence into I sets the level of every hit.
- **Bouncing LFOs**: channel 1 in LFO mode; its EOC into J fires channel 2 once per cycle.
- **Portamento**: a pitch sequence into H, attack = decay, and take the output with E fully clockwise.

6. SPECIFICATIONS

ITEM	VALUE
Width	10 HP
Channels	2
Attack / decay	0.3 ms – ~40 s, CV: +1 V halves the time
Shape	exponential – linear – logarithmic, continuous
Modes	one-shot, LFO (looping)
VCA input	normalled to 5 V; output = envelope × VCA × 2 × level
Outputs	±10 V, EOC 10 V / 0.5 ms, sum
Retrigger	ignored during attack, honoured during decay

CREDITS

Design, DSP and panel: betobullet. Behaviour modelled on the Cosmotronic Delta-V user manual; Delta-V is a product of Cosmotronic and this is an independent recreation. Fonts: Black Ops One, Space Mono, Space Grotesk (SIL OFL).